

A Technical Introduction to BLEU score

Section 1

r is the effective reference corpus length, that is, $r := \sum_{i=1}^M |y(i, j)|$ where $y(i, j) = \arg \min_{y \in S_i} |y| - |y \wedge \hat{y}|$, that is, the sentence from S_i whose length is as close to $|\hat{y}|$ as possible.

Section 2

Performance BLEU has frequently been reported as correlating well with human judgement,[10][11][12] and remains a benchmark for the assessment of any new evaluation metric. There are however a number of criticisms that have been voiced. It has been noted that, although in principle capable of evaluating translations of any language, BLEU cannot, in its present form, deal with languages lacking word boundaries.[13] Designed to be used for several reference translation, in practice it's used with only the single one. BLEU is infamously dependent on the tokenization technique, and scores achieved with different ones are incomparable (which is often overlooked); in order to improve reproducibility and comparability, SacreBLEU variant was designed. It has been argued that although BLEU has significant advantages, there is no guarantee that an increase in BLEU score is an indicator of improved translation quality.[14]

Section 3

Basic setup A basic, first attempt at defining the BLEU score would take two arguments: a candidate string $\hat{y}$ and a list of reference strings $(y(1), \dots, y(N))$. The idea is that $\text{BLEU}(\hat{y}; y(1), \dots, y(N))$ should be close to 1 when $\hat{y}$ is similar to $y(1), \dots, y(N)$, and close to 0 if not. As an analogy, the BLEU score is like a language teacher trying to score the quality of a student translation $\hat{y}$ by checking how closely it follows the reference answers $y(1), \dots, y(N)$. Since in natural language processing, one should evaluate a large set of candidate strings, one must generalize the BLEU score to the case where one has a list of M candidate strings (called a "corpus") $(\hat{y}^1, \dots, \hat{y}^M)$, and for each candidate string $\hat{y}^i$, a list of reference candidate strings $S_i := (y(i, 1), \dots, y(i, N_i))$. Given any string $y = y_1 y_2 \dots y_K$, and any integer $n \geq 1$, we define the set of its n -grams to be $G_n(y) = \{y_1 \dots y_n, y_2 \dots y_{n+1}, \dots, y_{K-n+1} \dots y_K\}$. Note that it is a set of unique elements, not a multiset allowing redundant elements, so that, for example, $G_2(abab) = \{ab, ba\}$. Given any two strings s, y , define the substring count $C(s, y)$ to be the number of appearances of s as a substring of y . For example, $C(ab, abcbab) = 2$. Now, fix a candidate corpus $S := (\hat{y}^1, \dots, \hat{y}^M)$, and reference candidate corpus $S = (S_1, \dots, S_M)$, where each $S_i := (y(i, 1), \dots, y(i, N_i))$. $S_i := (y^1(i, 1), \dots, y^{N_i}(i, N_i))$.

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Section 4

BLEU (bilingual evaluation understudy) is an algorithm for evaluating the quality of text which has been machine-translated from one natural language to another. Quality is considered to be the correspondence between a machine's output and that of a human: "the closer a machine translation is to a professional human translation, the better it is" – this is the central idea behind BLEU.[1] Invented at IBM in 2001, BLEU was one of the first metrics to claim a high correlation with human judgements of quality,[2][3] and remains one of the most popular automated and inexpensive metrics. Scores are calculated for individual translated segments—generally sentences—by comparing them with a set of good quality reference translations. Those scores are then averaged over the whole corpus to reach an estimate of the translation's overall quality. Intelligibility or grammatical correctness are not taken into account.[4] BLEU's output is always a number between 0 and 1. This value indicates how similar the candidate text is to the reference texts, with values closer to 1 representing more similar texts. Few human translations will attain a score of 1, since this would indicate that the candidate is identical to one of the reference translations. For this reason, it is not necessary to attain a score of 1. Because there are more opportunities to match, adding additional reference translations will increase the BLEU score.[5]

Section 5

c is the length of the candidate corpus, that is, $c := \sum_{i=1}^M |\hat{y}^i|$ where $|\hat{y}^i|$ is the length of $\hat{y}^i$.